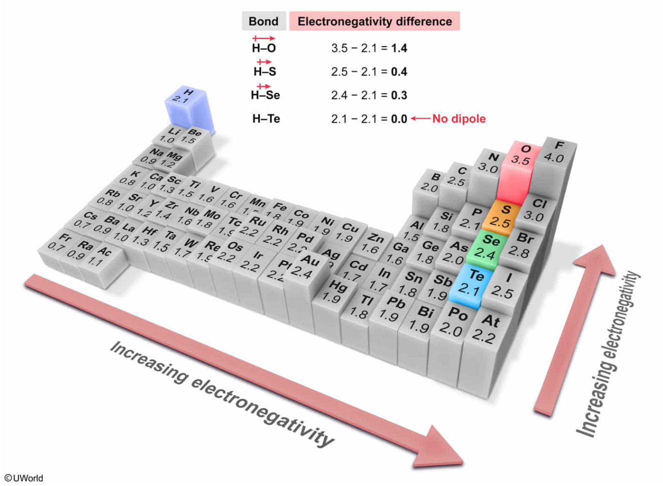
Which compound has the bond with the smallest dipole moment?

- A. H_2O [21%]
- B. H_2S [3%]
- C. H_2Se [1%]
- ✔ D. H_2Te [73%]

Correct
72% answered correctly

Explanation:



A covalent bond is formed by valence electrons shared between two atoms (usually two nonmetals). However, differences in electronegativity can result in unequal electron sharing. In such an arrangement, a polar covalent bond results, wherein the electrons in the bond are pulled closer to the atom of greater electronegativity (giving it a partial negative charge) and away from the atom of lesser electronegativity (giving it a partial positive charge).

The separation of partial charges across a polar covalent bond results in a vector quantity called a dipole moment. The magnitude of a dipole moment μ across a polar bond can be approximated by:

$$\mu = qr$$

where q is the magnitude of the partial charge separated across a distance r. The **magnitude** of the **partial charge** tends to increase as the **difference in electronegativity** between the two atoms in the bond increases, but the bond length across which the charge is separated also impacts the dipole.

The compounds H_2O, H_2S, H_2Se, and H_2Te each contain bonds with hydrogen. As such, the difference among these compounds is which of the Group 16 elements (ie, chalcogens) is bonded to hydrogen. Periodic trends in atomic radii show that Te is the largest chalcogen, which makes the H–Te bond the longest bond in the given compounds. However, the electronegativity difference is the smallest in the H−Te bond because electronegativity tends to decrease moving down a group (column) on the periodic table.

Therefore, the **H−Te bond is the longest** but has the smallest dipole moment of the given compounds because the atoms in the H−Te bond have almost no difference in electronegativity (ie, **negligible partial charge** separated across the bond).

(Choices A, B, and C) Although the hydrogen-chalcogen bond length decreases moving up a column on the periodic table, the electronegativity difference between the atoms increases significantly. As a result, the dipole moment of the hydrogen-chalcogen bond increases moving up the column. Therefore, the ranking of the dipole moments for each of the four bonds (from largest to smallest) is: H−O > H−S > H−Se > H−Te.

Educational objective:
The magnitude of a dipole moment across a polar covalent bond is equal to the magnitude of the partial charge multiplied by the bond length separating the charge. The difference in electronegativity between two covalently bonded atoms is indicative of the magnitude of the partial charge between bonded atoms. Periodic trends in atomic radii allow relative comparisons of bond length.

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